

SOFTWARE ENGINEERING LAB ASSIGN-1

Name and ID

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Problem Statement –

Student Mental Health Monitoring System

SDGs Covered-

- SDG 3 – Good Health and Well-being
- SDG 4 – Quality Education

Problem Statement

Mental health issues among students have become increasingly common due to academic pressure, competitive environments, social expectations, and lifestyle changes. Conditions such as stress, anxiety, depression, and burnout often go unnoticed until they significantly impact academic performance and personal well-being.

Traditional mental health support systems in educational institutions rely heavily on self-reporting or manual counseling processes. Many students hesitate to seek help due to stigma, lack of awareness, or limited access to counselors. As a result, early warning signs of mental distress are often missed.

The absence of proactive monitoring and early detection mechanisms prevents institutions from providing timely support. This can lead to poor academic outcomes, dropouts, and severe mental health crises. Hence, there is a need for an intelligent, data-driven system to monitor student mental well-being and enable early intervention.

Proposed System-

The **Student Mental Health Monitoring System** is an intelligent digital platform

designed to continuously assess and monitor the mental well-being of students using data analytics.

The system collects data from multiple sources such as

- daily study hours,
- sleep hours,
- screen time,
- academic performance,
- attendance records,
- behavioral patterns,
- periodic self-assessment questionnaires.

Algorithms analyze these inputs to identify patterns that may indicate mental stress, anxiety, or depression.

Based on the analysis, the system classifies students into different risk levels and generates alerts for counselors or authorized staff when intervention is required. A centralized dashboard allows institutions to monitor trends while maintaining student privacy and data security.

Objectives-

- To monitor student mental health using data analysis
- To detect early signs of stress, anxiety, and depression
- To enable timely intervention by counselors and institutions
- To reduce student dropouts and academic decline
- To promote mental well-being and emotional resilience
- To support a healthier educational environment

Conclusion

The Student Mental Health Monitoring System provides an effective and proactive approach to identifying and addressing mental health challenges among students. By utilizing data to analyze behavioral patterns, academic performance, and self-assessment responses, the system enables early detection of mental health risks that may otherwise go unnoticed.